

Blood glucose: build the control model

Teacher guide and worked answers

Explain the insulin response to raised blood glucose using a qualitative, controllable model.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx | LAUNCH Weekly - Summer | C31

Model blood-glucose control (interactive software).

Directly develops C31 and Pearson7.13. Glucagon under7.14 is clearly labelled optional Higher-tier stretch and kept separate from the core interaction. No diagnosis, blood-glucose thresholds, treatment choices or dose calculations are included.

Prior learning

Hormones travel in blood to responsive tissues.

Glucose can be used in respiration.

Negative feedback reduces a departure from a regulated condition.

Success evidence

Sequence raised glucose → insulin release → target response → movement towards a suitable range.

Distinguish glucose, glycogen and insulin.

Explain why a qualitative model cannot predict a person's readings or treatment.

Equipment

Shared evidence sheet, one selected response route, pen and ruler.

Projector or offline HTML/PowerPoint; calculator where useful.

Preparation

Print the shared page, one selected response route and the exit page. Routes are alternatives, not extra tasks.

Read the worked answers and note any constructed data before teaching.

Use the core interaction first. Keep the Higher glucagon panel closed unless that extension is appropriate.

All displayed levels are qualitative labels. Do not substitute pupil readings, medicine amounts or clinical thresholds into this classroom model.

Practical care

Paper and screen lesson; no human sampling, exercise test, breath holding, medicine handling or personal health disclosure.

Ready to teach alternative

All evidence is supplied. Use paper diagrams and cards, pointing, annotation or dictation if a device is unavailable.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Build a model that explains a direction
2	Retrieval	2-5	Message, material or store?
3	I do	5-9	Trace the core response
4	I do	9-12	Follow the material separately from the signal
5	I do	12-15	A direction model has a clear limit
6	We do	15-18	Control-panel challenge
7	We do	18-20	Repair the substance swap
8	We do	20-22	Test what the model cannot claim
9	Hinge	22-25	Which response belongs in the core model?
10	You do	25-26	Create the explanation that sits behind the controls
11	You do	26-36	Predict, explain and state a limit
12	Review	36-38	Audience: which arrow is the message?
13	Review	38-39	An accurate model is deliberately bounded
14	Exit	39-40	Keep the roles distinct
15	Exit	Within stage	Next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Build a model that explains a direction

Invite prediction; reassure that no personal diet, health or medication disclosure is required.

2 Retrieval Message, material or store?

R1: blood. R2: respiration and building/storing materials. R3: the original departure. Introduce glycogen as a store if the word is unfamiliar.

3 I do Trace the core response

Think aloud through the four-box model. Some body cells, such as muscle cells, increase uptake; liver and muscle storage as glycogen is promoted. Avoid saying every cell's uptake always requires insulin.

4 I do Follow the material separately from the signal

Trace purple signal arrow and orange material arrow separately. This is a molecule-role model, not a detailed chemical reaction mechanism.

5 I do A direction model has a clear limit

Model: "Later is nearer the usual range in this conceptual trace. I cannot read a medical value, a time-to-recovery or treatment from it." As glucose returns towards usual levels, the stimulus for extra insulin release reduces.

6 We do Control-panel challenge

Use manual stage controls. Start with usual, then absorption/raised, insulin response, target response and towards usual. Paper alternative orders cards A-D.

7 We do Repair the substance swap

Glucose units are stored as glycogen; insulin signals target processes. Ask which arrow carries the message and which carries material.

8 We do Test what the model cannot claim

No: no calibrated concentrations, times, personal physiology or meal quantities. It is designed for mechanism and direction only.

9 Hinge Which response belongs in the core model?

First option correct. If pupil swaps insulin/glucose, return to the two-arrow diagram.

A. Raised glucose → more insulin release → uptake/storage responses. - This is the core corrective pathway for raised blood glucose.

B. Insulin becomes glucose, so blood glucose rises further. - Insulin is a signal, not a sugar product.

C. The picture supplies an exact medicine dose. - There are no clinical measurements or dose rules in this model.

10 You do Create the explanation that sits behind the controls

Keep the Higher branch optional and visually distinct. Do not require it for core success.

11 You do Predict, explain and state a limit

Supported S1-S4; standard M1-M4; stretch T1-T4. T3 is optional Higher glucagon; other evaluation is not a tier label.

12 Review Audience: which arrow is the message?

Pupil traces or labels the two arrows; quiet partner/TA review is acceptable.

13 Review An accurate model is deliberately bounded

Restate that qualitative usual/raised labels are not clinical categories for any pupil.

14 Exit Keep the roles distinct

E1: pancreas. E2: glucose units. E3: no calibrated clinical or individual information. Re-teach substance roles first if needed.

15 Exit Next useful step

Shared exit minute. Accept a labelled route and a spoken limit.

Answers and assessment

Retrieval and We do

R1: blood; R2: respiration and material use/storage; R3: the original departure. W1: $A \rightarrow B \rightarrow C \rightarrow D$. W2: glucose units, not insulin, are stored as glycogen; insulin signals the processes. W3: no exact concentrations, times or individual parameters are supplied.

Hinge

Correct: raised glucose stimulates more insulin release and uptake/storage responses. Insulin is not converted into glucose. The model supplies no dosing information.

S1 / M1

After absorption raises blood glucose, pancreatic beta cells release more insulin into blood. Insulin-responsive tissues alter glucose handling; some body cells increase uptake and liver/muscle glycogen storage is promoted.

S2 / M2

Insulin=chemical messenger hormone; glucose=sugar/material; glycogen=store of joined glucose units. A hormone signal changes cell processes; it is not the bulk material being stored.

S3

Uptake by insulin-responsive tissues such as muscle and storage as glycogen in liver/muscle help remove glucose from blood, tending to reduce the raised concentration.

S4 / M3

As glucose moves towards the suitable range, the stimulus for extra insulin release reduces. The response counteracts the original rise rather than amplifying it indefinitely.

M4 / T2

No. The model has no calibrated concentrations, real time scale, individual physiology, absorption details or measured response parameters. Similar qualitative patterns need not have identical numerical values or timing.

T1

Some tissues take up glucose through insulin-independent routes. Better: insulin promotes uptake/storage responses in particular tissues, including increased uptake in muscle cells; it does not control every uptake route in every cell.

T3 optional Higher

Lower glucose stimulates pancreatic alpha cells to release glucagon. Glucagon promotes liver glucose release, including glycogen breakdown, tending to raise blood glucose towards usual levels. Insulin responds to raised glucose and promotes uptake/storage; both reduce their original departure. Core success does not require glucagon.

T4

A time/concentration axis with defined units could support comparing measurements. The relationships and parameters would still need testing against independent measured data; adding an axis alone does not show the model predicts accurately for different individuals.

Exit and assessment

E1: pancreas. E2: glucose units. E3: no calibrated concentration scale, real time scale or individual response parameters. L1: specific mechanism correction or justified confirmation. Secure core: correct insulin sequence, substance distinction and model limit.

Teaching and assessment notes

40-minute lesson. Zero-minute slides share the preceding stage allocation.

Choose one response route. Accept accurate speech, pointing with explanation, annotations or writing; record the support used.

A teacher or TA can be a quiet audience: ask one evidence question, then let the pupil revise or justify a decision.

Core control panel remains qualitative and contains no blood-glucose values or doses. Advance only on a pupil or teacher action.

The Higher glucagon extension is separate and closed by default. It is never required to answer core insulin questions.

The paper route orders and explains the same A-D events. The animation represents an authored causal sequence, not a numerical physiology simulation.

Access and responsive teaching

Supported

Same core concepts with chunked prompts and read-aloud access. Record support used; a pathway is not a diagnosis or fixed attainment label.

Standard

Explain the supplied evidence independently using accurate scientific terms and units.

Stretch

Optional deeper transfer or evaluation. Higher-tier-only material is labelled explicitly and is not required for the core outcome.

Regulation

Offer solo or partner work, a quiet response, a pass and brief reset with clear re-entry. No forced disclosure or public performance.

Misconceptions to check

Insulin is the same substance as glucose.

Insulin is a hormone; glucose is a small sugar molecule.

Check: Ask which is the message and which is a material used by cells.

Insulin itself turns into glycogen.

Insulin signals target processes; glucose units can be stored as glycogen.

Check: Trace the hormone arrow separately from the glucose-material arrow.

Every cell needs insulin for every glucose uptake process.

Insulin affects particular target tissues; several tissues also take up glucose through insulin-independent routes.

Check: Use “some body cells, such as muscle cells” in the core explanation.

The model can tell someone their required insulin dose.

It has no clinical measurements, individual parameters or treatment rules.

Check: Ask which real-world information is deliberately absent.

Word	Meaning
blood glucose concentration	The amount of glucose in a stated volume of blood.
insulin	A hormone released by pancreatic beta cells that helps lower raised blood glucose through target-tissue responses.
glycogen	A stored carbohydrate made from joined glucose units, especially in liver and muscle.
pancreas	An organ with endocrine cells that release hormones including insulin.
negative feedback	A response that reduces the original departure from suitable conditions.
qualitative model	A model showing the kind or direction of a change rather than calibrated numerical predictions.

Scientific references

[Pearson Edexcel GCSE Biology 1BI0 specification](#)

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 9 September 2026.

[NIDDK: What is diabetes?](#)

Checked pancreas/insulin and uptake mechanism. The lesson is about normal control concepts, not diagnosing diabetes. Checked 9 September 2026.

[OpenStax Anatomy and Physiology2e: The endocrine pancreas](#)

Checked insulin, glycogen storage and the separately labelled glucagon extension. Checked 9 September 2026.